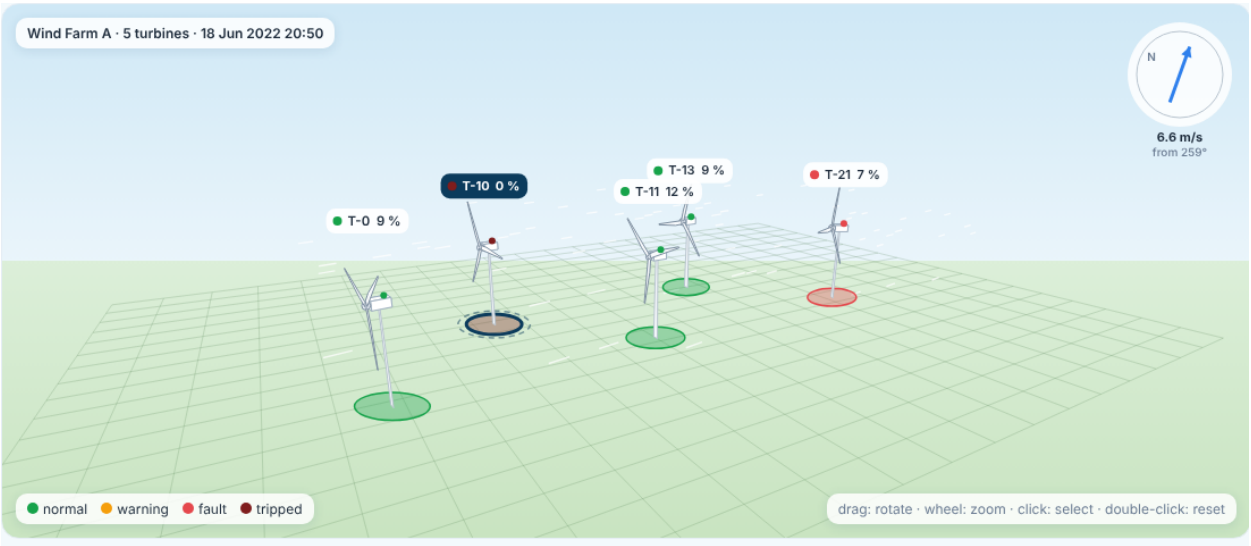
Wind Farm Digital Twin

User manual

Power forecasting, anomaly detection and classification, a live 3D simulation of 5 turbines with anomaly injection, and a results dashboard - built on the CARE to Compare SCADA dataset (Wind Farm A).



The 3D wind farm view of the app (rendered by the app's own 3D component from a live simulation): T-10 stopped by its protection after an injected severe gearbox fault, T-21 in fault with a yaw misalignment (rotor turned away from the wind).

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1 What is in the bundle

Folder / file	Contents
digital_twin/	the app (app.py), simulator and live twin (twin_core/), 3D view (components/farm3d/), turbine models (models/, created on the first start)
forecasting_model/	power_model.py (training), evaluate_forecasting.py (MAE, RMSE, R2, MAPE + figures), trained model in outputs_power_model/, results/
anomaly_detection_model/	anomaly_classifier.py (Step 1 comparison), anomaly_classifier_v2.py (attempt 2), anomaly_classifier.joblib (Random Forest used by the app), per-class metrics and confusion matrices in results_attempt1/ and results_attempt2/, fault_detection*.py (normal-behaviour models)
dashboard/	results_dashboard.py (also a page of the app), export_pdf.py, results_dashboard.pdf
data/	processed/ (extracts used to train the classifier) and README.md (where to put the raw dataset)
requirements.txt, README.md	packages and quick start
docs/	this manual's source (make_manual.py) and screenshots

All paths are relative to the bundle folder, so the project runs from any location on Windows, macOS or Linux.

2 Installation

2.1 Python and packages

Python 3.10 - 3.13 (Anaconda / Miniconda or python.org). No GPU or special hardware is needed. In a terminal (Windows: **Anaconda Prompt**), go to the bundle folder and install:

```
cd path/to/RET_Wind_Farm_Twin
pip install -r requirements.txt
```

With conda you can use your existing environment:

```
conda activate ret
pip install -r requirements.txt
```

2.2 The raw dataset (once)

The first start builds the 5 simulated turbines from the real data. Extract the CARE to Compare zip (Zenodo record 14958989) so that a folder **Wind Farm A** (with event_info.csv and the dataset CSV files) is either inside **data/** or in the folder that **contains** the bundle (for example your old project folder). The turbines imitate the healthy datasets 24, 3, 25, 14 and 13 (assets 0, 10, 11, 13 and 21).

3 Running each part

Part	Command (from the bundle folder)	Notes
Digital twin app	streamlit run digital_twin/app.py	opens http://localhost:8501; first start 5-8 min (turbines are built once)
Results dashboard alone	streamlit run dashboard/results_dashboard.py	static results, PDF export
PDF report	python dashboard/export_pdf.py	writes dashboard/results_dashboard.pdf
Forecast evaluation	python forecasting_model/evaluate_forecasting.py	MAE, RMSE, R2, MAPE, figures
Prepare turbines manually	python digital_twin/twin_core/prepare.py	same as the first start of the app
This manual	python docs/make_manual.py	adds docs/screenshots/app_*.png

Stop the app with **Ctrl + C** in the terminal.

4 The screens

4.1 Sidebar (always visible)

• **Navigation** - the five pages.

• **Play** - runs the simulation; every second (about) the farm advances.

• **Speed** - 10-minute steps per second (1, 3, 6, 12 or 36 = 6 hours per second).

• **Weather** - normal, windy, calm, storm (very high wind, cut-out), hot day, cold day.

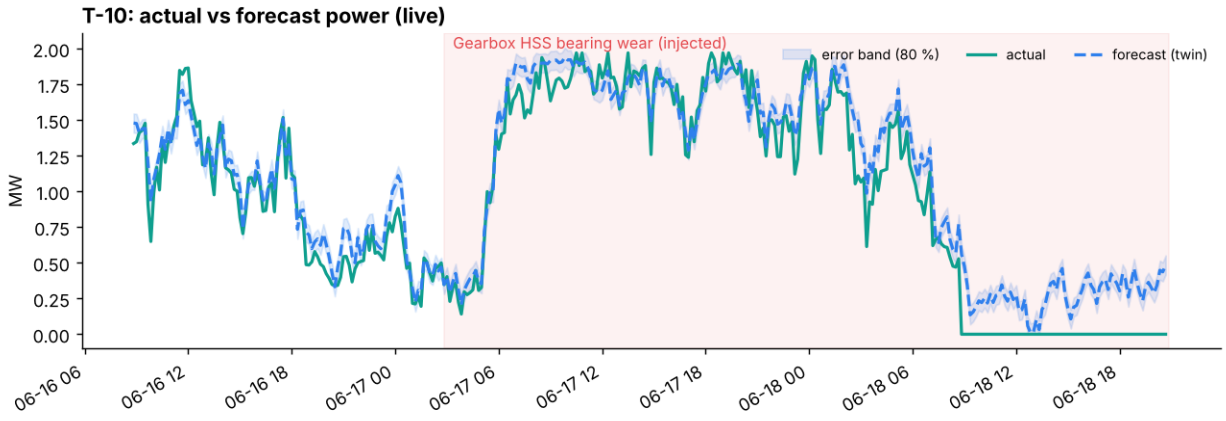
• **Rated power** - MW per turbine (the dataset power is scaled 0-1; used for MW and MWh).

• **Status colours from** - which detector decides normal / warning / fault: the Random Forest (Step 1 model C, default), the residual detector + energy monitor, or either.

• **+6 h** jumps ahead, **Reset** starts a new simulation.

4.2 Wind farm (live)

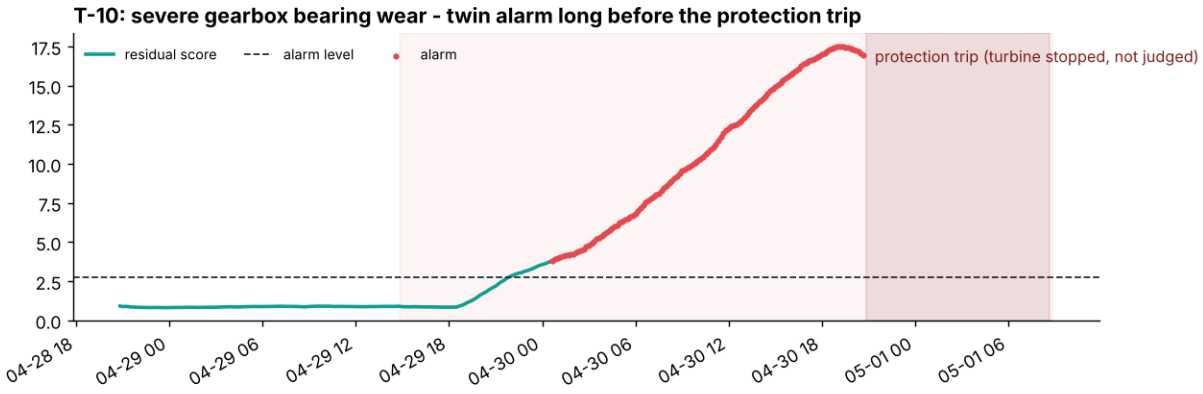
Top: simulated time, farm power now (actual and forecast), energy of the session, number of turbines in fault. Left: the **3D view** - rotors spin at the simulated rotor speed, each nacelle points where the turbine is yawed, the compass shows where the wind comes from; the ring and lamp colour show the status (green normal, amber warning, red fault, dark red tripped). **Drag** to rotate, **wheel** to zoom, **click** a turbine to select it, **double-click** to reset the camera. Right: live readings of the selected turbine - wind, direction, yaw error, ambient, rotor / generator speed, pitch, actual and forecast power, gearbox, generator bearing, stator, transformer, hydraulic and converter temperatures, plus the Random Forest class and the residual check. Below: actual vs forecast power with the 80 % error band (injected anomalies shaded red), component temperatures, and the stacked farm power.



Actual vs forecast power of the selected turbine with the error band (live chart; same data as in the app).

4.3 Anomalies

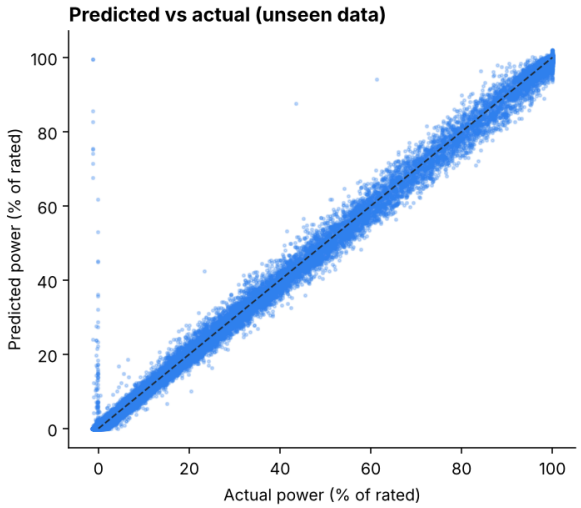
Anomaly injection panel (section 5). **Current state of every turbine**: status, Random Forest class and confidence, its alarm, the residual component and score, the energy performance index, timestamp, likely cause and the injected anomaly (if any). **Injected anomalies**: for each injection the class / component found, the detection delay, energy lost and whether the turbine tripped; an active one can be repaired. **Detection log**: every alarm with time, detector, result, confidence, the injection it belongs to (or *none - false alarm?*) and the likely cause. Chart: the detector signals of the selected turbine.



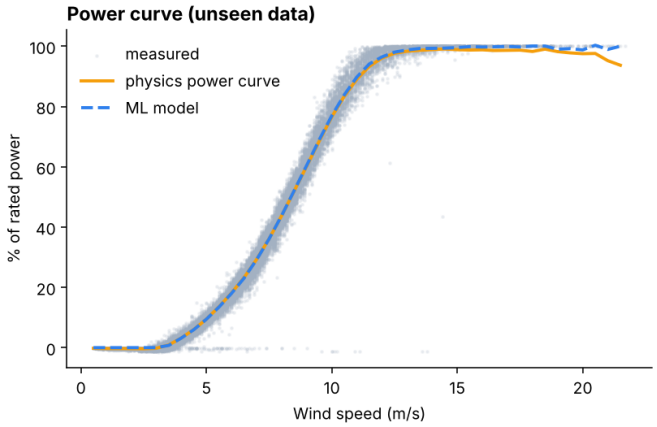
Detector signal of a turbine with an injected severe gearbox fault: the twin's alarm comes about 20 hours before the turbine's protection stops it.

4.4 Forecasting details

A step-by-step explanation for the presentation and viva: (1) data and preprocessing, (2) features and their permutation importance, (3) model choice and why, (4) training / test results - MAE, RMSE, R2 and MAPE with the physics power curve for comparison and per turbine, (5) predicted vs actual, residual distribution, residual vs wind speed, power curve, (6) how to explain it.



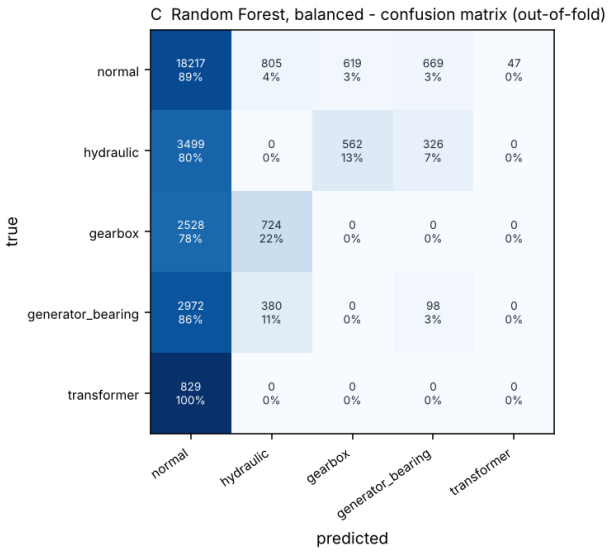
Predicted vs actual power on unseen data.



Power curve: measured, physics curve and ML model.

4.5 Results dashboard

Energy per turbine and for the farm (actual, expected, lost to injected anomalies), forecast metrics, the anomaly detection metrics on real data (before vs after table, per-class table and confusion matrix of the Random Forest, attempt 2), the injected anomalies with detection results, energy lost per anomaly, the anomaly history timeline, and **Build PDF report** (download). Every chart has a camera icon for a PNG.



Confusion matrix of the Random Forest on real data (out-of-fold, leave-one-turbine-out).

Results dashboard

Wind farm digital twin - results report

Data: CARE to Compare, Wind Farm A (5 turbines, 22 datasets, 12 recorded faults of 4 types).

FORECASTING (power from weather, unseen real data)

```
  MAE  1.15 % of rated
  RMSE 1.82 % of rated
  R2   0.995
  MAPE 4.8 % (rows above 10 % of rated)
```

ANOMALY DETECTION (real data, leave-one-turbine-out)

```
  Model in the app: Random Forest (Step 1, approach C), 175 features, class-balanced.
  Cross-check: residual detector (normal-behaviour models) + 24-h energy performance monito
  A  old model (v1 threshold + diagnosis)        detected  1/12  correct  0/12  false alarm
0/10
  B  Gradient Boosting, class-balanced           detected  9/12  correct  1/12  false alarm
7/10
  C  Random Forest, balanced                     detected  8/12  correct  1/12  false alarm
6/10
  D  residual detector + fault-type classifier   detected 10/12  correct  3/12  false alarm
8/10

  With 1-6 events per fault type the fault type cannot be classified reliably on unseen
turbines;
  even with settings tuned on all data at most 3 of 12 faults are classified correctly.
```

First page of the PDF export.

5 Injecting anomalies

Page **Anomalies** -> panel **Anomaly injection**: choose the turbine, one of the 12 anomaly types, the severity, how it develops and the duration, then press **Inject anomaly** and **Play** (or +6 h). The simulated turbine's signals change; the twin does not know about the injection and has to find it.

Anomaly	Component (residual)	Random Forest class	What changes
Gearbox HSS bearing wear	Gearbox	Gearbox	temperature rise: s11, s12
Gearbox oil overheating	Gearbox	Gearbox	temperature rise: s12, s11
Generator bearing damage (DE)	Generator bearings	Generator bearing	temperature rise: s13, s14
Generator bearing damage (NDE)	Generator bearings	Generator bearing	temperature rise: s14, s13
Transformer phase overheating	Transformer	Transformer	temperature rise: s39, s38
Transformer cooling failure	Transformer	Transformer	temperature rise: s38, s39, s40
Hydraulic oil overheating	Hydraulic group	Hydraulic group	temperature rise: s41; pitch offset
Hydraulic pressure loss (pitch lag)	Hub / pitch	Hydraulic group	temperature rise: s41, s6; pitch offset; power loss
Pitch misalignment	Hub / pitch	not a trained class	pitch offset; power loss
Yaw misalignment	Power performance	not a trained class	yaw offset (power x cos^3)
Converter cooling failure	Converter	not a trained class	temperature rise: s35, s36, s37, s21
Blade icing / soiling	Power performance	not a trained class	power loss

Setting	Meaning
Severity	Mild 5, Moderate 17, Severe 25 x the sensor's normal noise (temperatures); pitch 1.5 / 3 / 6 deg; yaw 8 / 15 / 30 deg; power loss 6 / 15 / 30 %. Temperature rises grow with the load (0.6 + 0.4 x load).
How it develops	Sudden, or growing linearly over 1 day / 3 days
Duration	6 - 120 h; afterwards the anomaly ends (repaired). 'Repair now' ends it at once.
Protection trip	a Severe temperature anomaly at full level for 6 h stops the turbine (rotor stops, 3D view dark red, energy lost); the twin pauses judging a stopped turbine
Energy lost	sum of (forecast x the turbine's healthy performance level - actual power) while the anomaly is active, x rated MW x 1/6 h

What to expect

• **Residual detector + energy monitor**: on the simulated turbines it found 11 of the 12 anomaly types with the correct component (moderate / severe), typically 4 - 25 hours after the start, and gives an early warning before a protection trip. Pitch misalignment is often missed (pitch is noisy). Power-loss anomalies need roughly > 15 - 20 % loss.

• **Random Forest (Step 1 model C)**: as on the real data, it mostly answers 'hydraulic group' and also flags healthy turbines; its class is shown but should not be trusted (section 6).

• Synthetic anomalies show how the twin reacts. They are not used to claim accuracy.

6 The models and their real-data results

6.1 Forecasting (power from weather)

Gradient Boosting on density-corrected wind, wind, gustiness, direction, temperature, air density, time of day and season; physics power curve (IEC 61400-12-1) as baseline. Unseen test data: **R2 0.995, RMSE 1.82 %, MAE 1.15 % of rated** (physics: R2 0.993, RMSE 2.12 %). MAPE (rows above 10 % of rated) from evaluate_forecasting.py.

6.2 Anomaly detection and classification (Step 1)

12 recorded faults of 4 types (hydraulic 6, gearbox 3, generator bearing 2, transformer 1) and 10 healthy datasets. Leave-one-turbine-out cross-validation, running turbine only, logbook status flags not used as inputs (they mark the fault windows - using them gives a fake 100 %).

Approach	Faults detected	Correct type	False alarms (healthy)
A old model (threshold + diagnosis)	1/12	0/12	0/10
B Gradient Boosting	9/12	1/12	7/10
C Random Forest (used in the app)	8/12	1/12	6/10
D detector + fault-type classifier	10/12	3/12	8/10
E fault-component detector + physics rule	2/12	1/12	1/10

Upper limit: even with settings tuned on all data, at most 3 of 12 faults can be classified correctly - most faults do not rise above the normal variation of healthy turbines in 10-minute SCADA data, and there are only 1 - 6 events per type. The twin therefore shows the residual detector + 24-h energy performance monitor next to the Random Forest.

7 Retraining the models

Model	Command	Time
Forecasting model	python forecasting_model/power_model.py	~2 min
Forecast evaluation	python forecasting_model/evaluate_forecasting.py	< 1 min
Normal-behaviour models (features)	python anomaly_detection_model/fault_detection.py python anomaly_detection_model/fault_detection_v2.py	15-20 min each
Data extracts for the classifier	python data/processed/extract_signals.py	~2 min
Random Forest - final model only	python anomaly_detection_model/anomaly_classifier.py --final-only	~1 min
Step 1 full comparison (CV)	python anomaly_detection_model/anomaly_classifier.py	~50 min
Attempt 2	python anomaly_detection_model/anomaly_classifier_v2.py	~5 min
Simulated turbines	python digital_twin/twin_core/prepare.py	1-2 min per turbine

If a saved model cannot be loaded with your scikit-learn version, the app rebuilds it automatically (Random Forest: final model only; turbines: prepare).

8 Troubleshooting

Problem	Fix
'streamlit' is not recognized	the environment is not active: conda activate ret (Anaconda Prompt), or use python -m streamlit run digital_twin/app.py
ModuleNotFoundError	pip install -r requirements.txt in the same environment
'Wind Farm A dataset was not found'	put the CARE_To_Compare folder into data/ or next to the bundle (data/README.md)
First start takes long	normal: 5 turbines are built from the real data once (5-8 min); later starts take seconds
Whole page reloads during Play / slow	update Streamlit to 1.37 or newer; use speed 36 for fast progress; close other heavy programs
3D view is empty	refresh the browser page (F5); the view needs a browser with canvas support (Chrome, Edge, Firefox)
Many sklearn UserWarning messages	harmless; they are hidden in the scripts
Port 8501 in use	streamlit run digital_twin/app.py --server.port 8502

9 Limitations

• The 'live' data is simulated from models trained on the real turbines; the turbines are not connected to a real plant.

• Fault types cannot be classified reliably from this data (section 6); accuracy is reported only on real data.

• Power in the dataset is scaled 0-1; MW and MWh use the rated power set in the sidebar (default 2 MW).

• The 3D view is a schematic visualisation (turbines drawn about twice their real size).